

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Software Engineering [IT_4061]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze the key changes that transformed traditional coding practices into modern software engineering principles. How did these changes influence project scalability? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply your understanding of software life-cycle models to demonstrate why iterative and incremental models are often preferred over the classical Waterfall model in dynamic business environments. (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Analyze how the Spiral model integrates elements of both design and prototyping. What risks does it aim to mitigate compared to other models? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Analyze the impact of poor stakeholder communication during the requirement-gathering phase. How does it affect downstream software quality? (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concepts of cohesion and coupling to demonstrate how they influence the modular structure and reusability of a software system. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Design a Data Flow Diagram (DFD) for a library management system to illustrate and justify how key functions and data movements are identified within the system. (5)
- Course Outcome: "CO3" / Cognitive Level: "Create"
- 4) Construct a comprehensive DFD for an online ticket-booking system to show the process of refining high-level diagrams into detailed components. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Create"
- B) Illustrate and analyze how applying design patterns simplifies recurring design problems and supports scalability in OO systems. (5)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Evaluate the effectiveness of using activity diagrams to represent system workflows during the early design phase. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Evaluate how integrating black-box, white-box, and fuzz testing enhances defect detection coverage in software products. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4061:Software Engineering

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4061:Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	4	20(40%)
5	Create	2	10(20%)
6	Evaluate	2	10(20%)
Total		10	50 (100%)